TANAYJYOT SINGH CHAWLA

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EDUCATION

University of Toronto – HBSc in **Computer Science** *Sept 2023 – Jun 2027*
Minor in Economics and Certificate in Business Fundamentals Annual GPA: 3.61

- **Co-President** (UTMIST): Led **North America's largest AI student organization** (350+ executives), secured $60k+, reached 500k+ monthly views, ran 5 events with 300+ attendees, 30+ total events, impacting 3500+ students.
- **National Coding League Finalist**: Ranked 3rd in 5000+ teams (25,000+ participants) solving Data Structures problems.

WORK EXPERIENCE

Shopify | MLOps Intern *May 2026 – Aug 2026*

- Owned the team's **H100** GPU cluster: benchmarked **vLLM** against **TensorRT-LLM** for small-model serving and moved latency-tolerant jobs to batch inference after tracing a **CentML** serving-layer bottleneck.
- Built an **Apache Airflow** pipeline that retrains the recommender, benchmarks it against the production model, and promotes the winner automatically; lifted Recall@200 by **39.5%** and increased Shop app click through rate.
- Redesigned the recommender benchmark pipeline (recall, leave-one-out, NDCG) that ranks every deployed model against shared ground truth, simplifying it into a self-serve **Tangle ML** tool to test new models against production.
- Registered **790** inference features in **Feast** with versioning and drift checks so **BigQuery** (offline) and **BigTable** (online) materialize from one definition, giving ML engineers self-serve features without depending on data-engineers.

Human-Centered Data Science Lab | ML Research Assistant (Submitted ACM CHI'27) *Jun 2026 – Present*

- Sovereign on-prem LLM infrastructure for the Children's Aid Society of Toronto, extending the **CHI '26 study** in public services, so **56K+** case notes are analyzed without leaving agency-controlled hardware.
- Built **rtxllm**, a **llama.cpp** serving CLI for the lab's RTX 5080: a 15-model quantized **GGUF** catalog, an OpenAI-compatible **REST** gateway, and zero-downtime model swaps with automatic rollback.
- Profiled a 23k-note sweep on 8 **Tenstorrent** chips (79% prefill-bound); chose tensor over data parallelism and removed a duplicated warmup prefill, cutting batch time **44%** with byte-identical output.
- Hit **406 tok/s** at batch 16 on Qwen3-14B; traced corrupt output to tt-metal kernel bug #35225 and dropped prefix caching.

RBC Borealis | Let's Solve It Fellow *Mar 2026 – May 2026*

- Engineered **Python/LangChain** pipelines condensing **1,100** pages into one-page digests, cutting review time by **90%**.

Flybits | AI Research Intern (Published MDPI Data '26) *Mar 2025 – Jan 2026*

- Built multi-agent **LangGraph/Pydantic** pipelines that simulate how a person's or company's state evolves over time, with schema-validated state, persistent memory, and validation-gated transitions with bounded retries.
- Designed a 5-entity, 50-relation knowledge-graph schema for business events; co-authored the resulting **PAiNT paper** and 1,200-artifact benchmark with **MIT Media Lab (sAIpien)**, published in *Data* (2026).

PROJECTS

AI Warehouse Battle Simulation (Published CUCAI '25) *Sep 2024 – May 2025*

- Led the UTMIST × AI Warehouse (**500k+** subs) team building a Unity battle sim where two species of up to **100** agents each fight, reproduce and evolve against each other in real time via a **C# NEAT** package.
- Built XOR and sine-fitting benchmarks with a custom fitness function (MSE, max error, complexity penalty), tuned evolution hyperparameters, and a **Graphviz** pipeline that snapshots network topology at every improvement.

SKILLS

Languages: Python, C/C++, Bash, SQL, TypeScript, Java, C#
ML & Inference: PyTorch, HuggingFace, scikit-learn, llama.cpp, vLLM, TensorRT-LLM, LangGraph, Pydantic, Feast
Infra & Tools: Docker, Kubernetes, GCP (BigQuery/BigTable), Apache Airflow, Tangle ML, Skypilot, Tailscale, Git